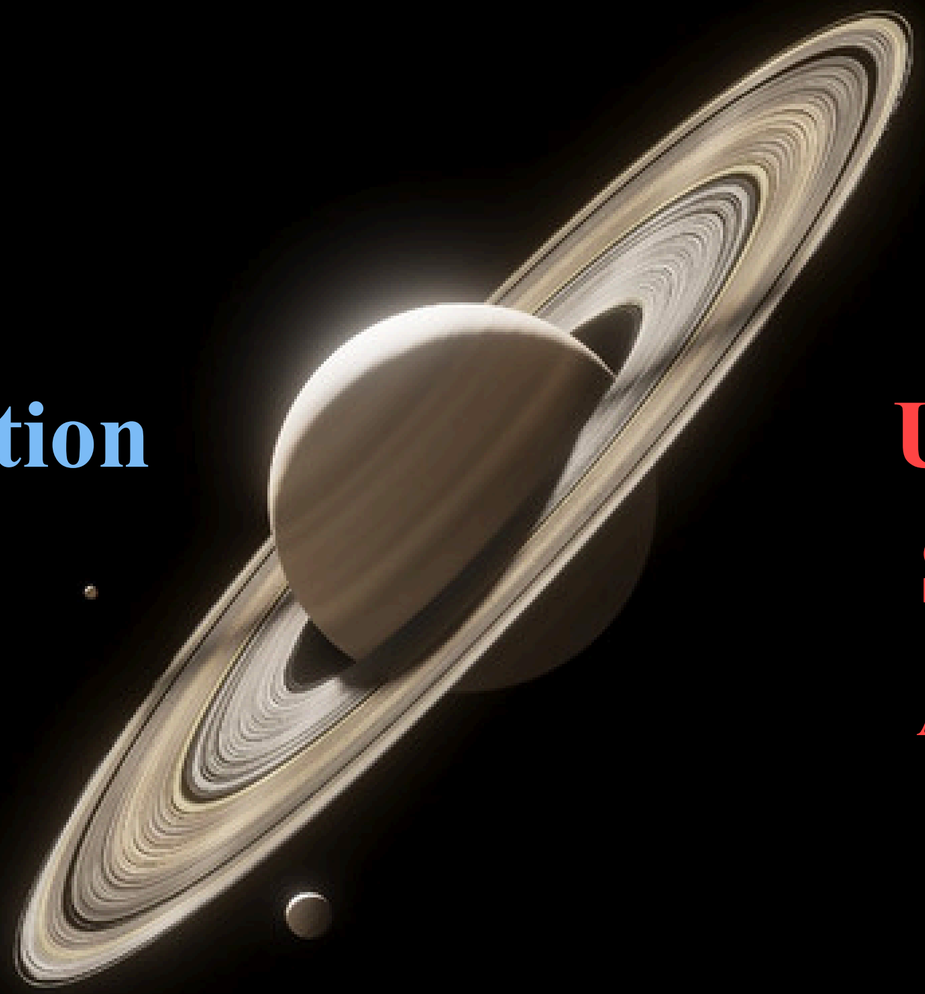
Final Presentation - Group S3

Planetary Rotation of Saturn



Using Doppler Spectroscopy

Advisor: Mai Nhu Tin

IAU

ISYA
INTERNATIONAL SCHOOL FOR YOUNG ASTRONOMERS

Det Norske Videnskaps-Akademi
The Norwegian Academy of Science and Letters

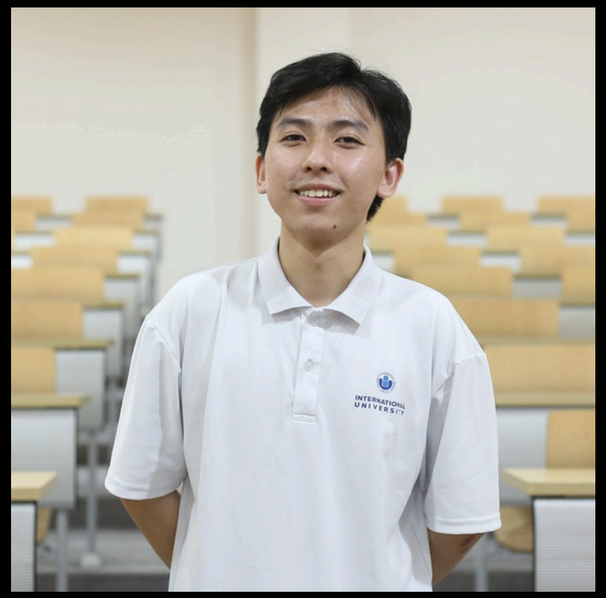
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RENCONTRES DU VIETNAM

SAGI

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Our group



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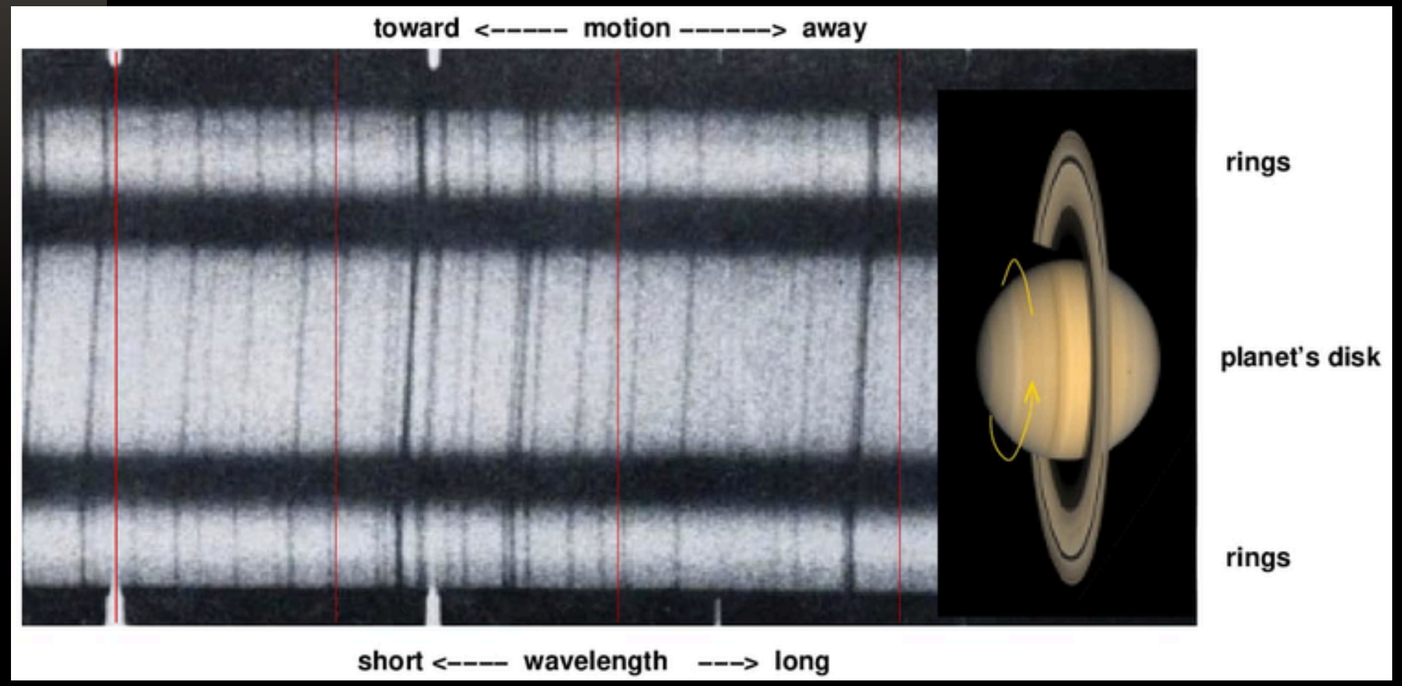
Workflow

Data & Methods

Results & Discussions

Conclusion

INTRODUCTION



From Gingerich, Sky and Telescope, Nov 1964, p.278

Primary objective

Saturn's equatorial rotation velocity by analysing Doppler shifts in its absorption spectral lines.

WORKFLOW

Data Acquisition at QNO

Literature, Method Review

Data Reduction with Python

Data Reduction with Demetra

Wavelength Calibration with Demetra

Data Analysis with Python

Final Report

Final Presentation

WEEK 1 **WEEK 2** **WEEK 3**



WORKFLOW

Data Acquisition at QNO

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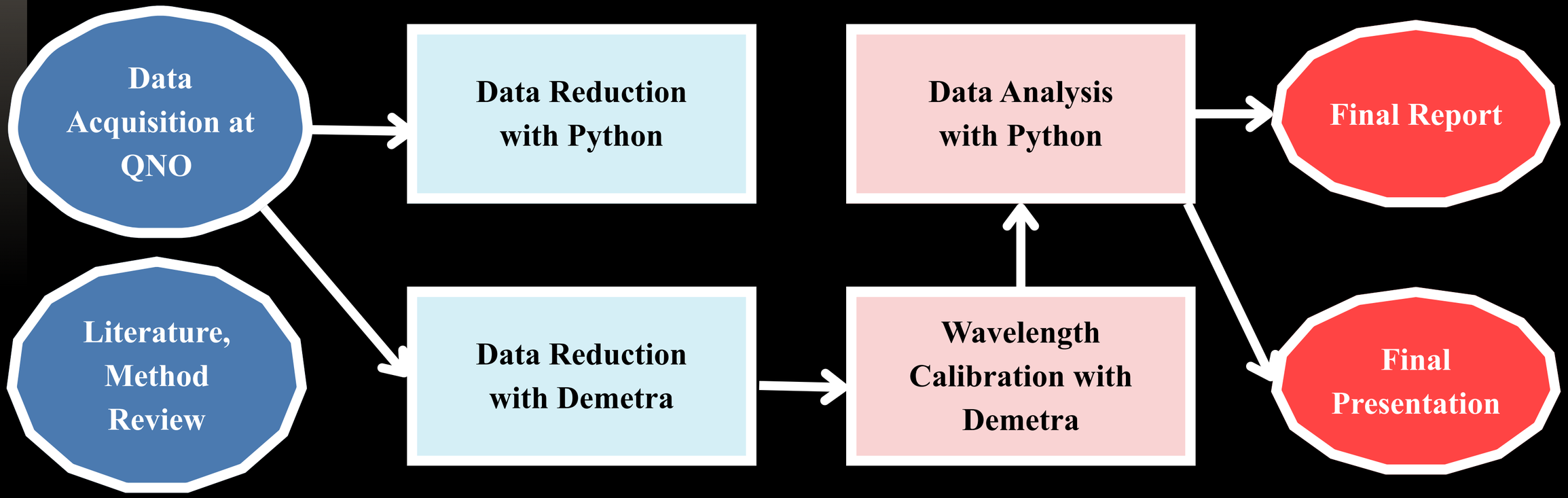
Final Report

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WEEK 1 **WEEK 2** **WEEK 3**



WORKFLOW



WEEK 1 **WEEK 2** **WEEK 3**



DATA & METHODS

Instruments

CDK600 Telescope System



Aperture: 24-inch
Focal length: 3974mm
A altazimuth mount

Spectrometer



Littrow configuration
Slit narrow: 15μm
Grating: 2400l/mm

Calibration Module



Calibration by Argon-Neon light.

Flat-field calibration by
Tungsten(halogen) light

ASI183mm Camera

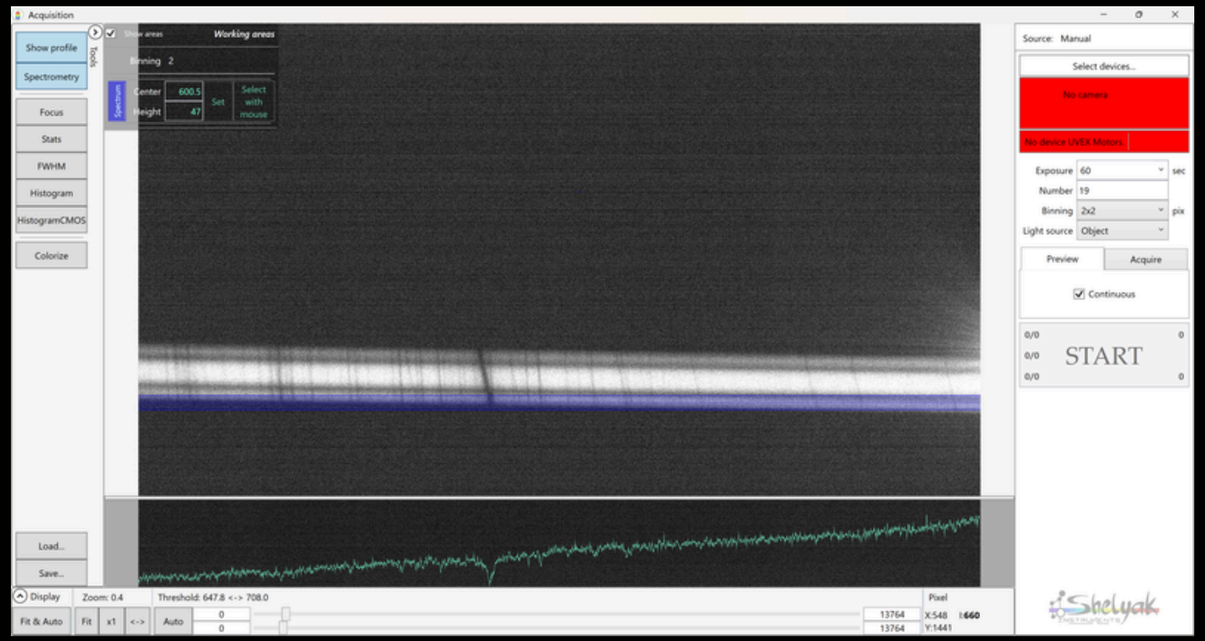


Wavelength Range: 0.4 μm to 0.75 μm (over-50% relative response)

Sensor Size: 1"

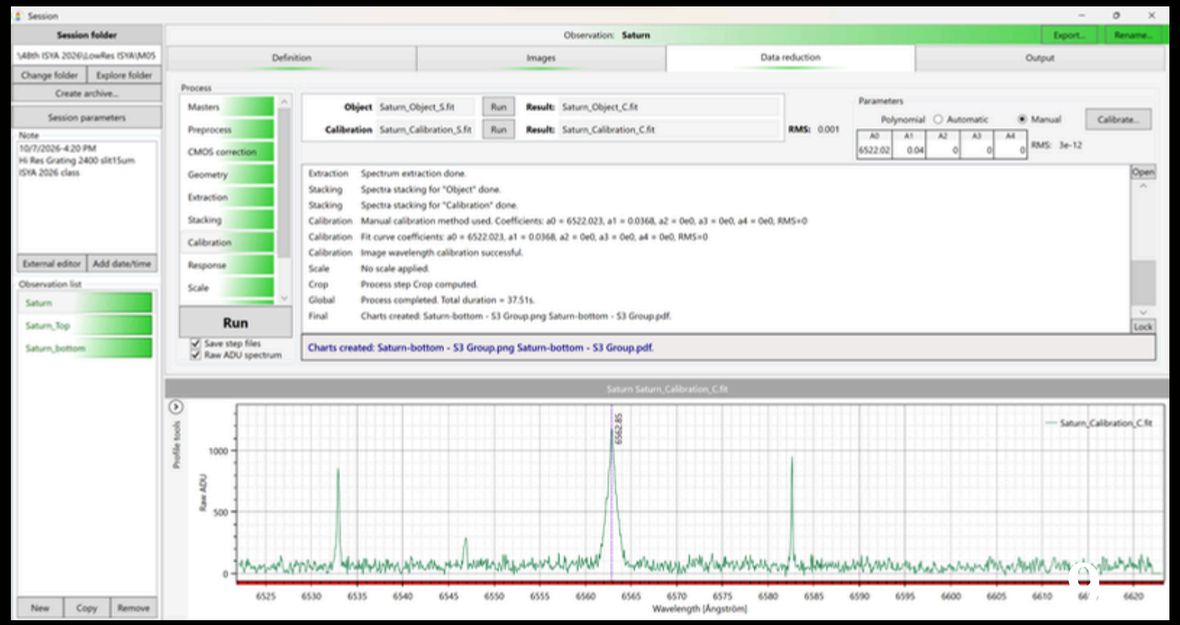
DATA & METHODS

Demetra Software



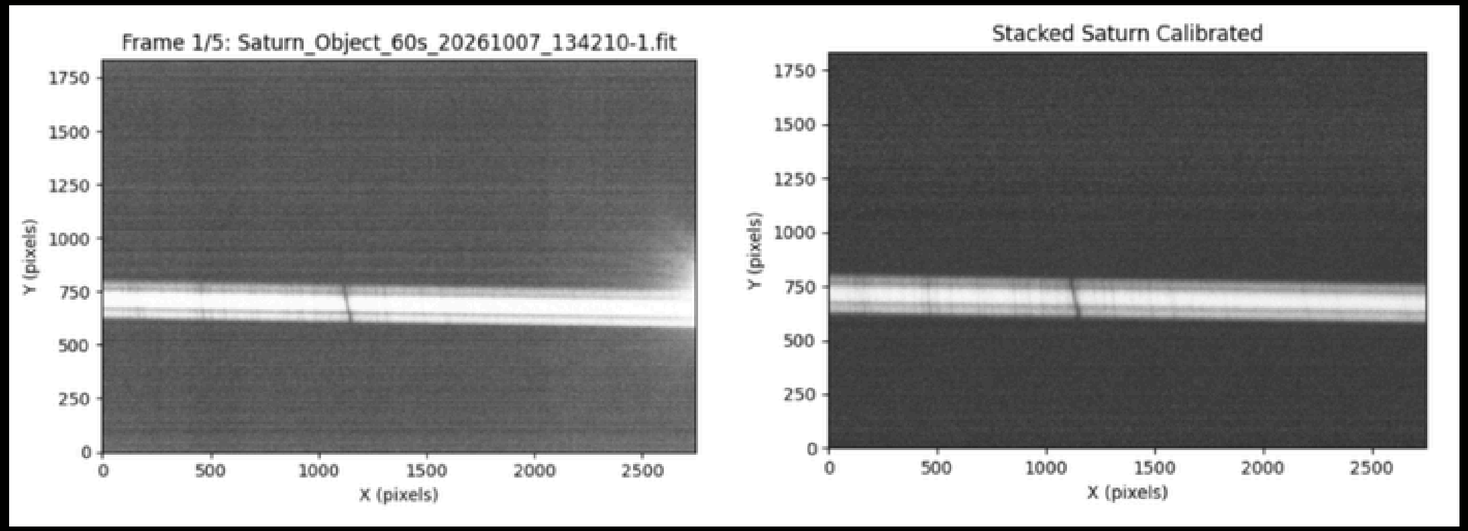
Acquisition Interface of the Demetra Software

Data Reduction Interface of the Demetra Software

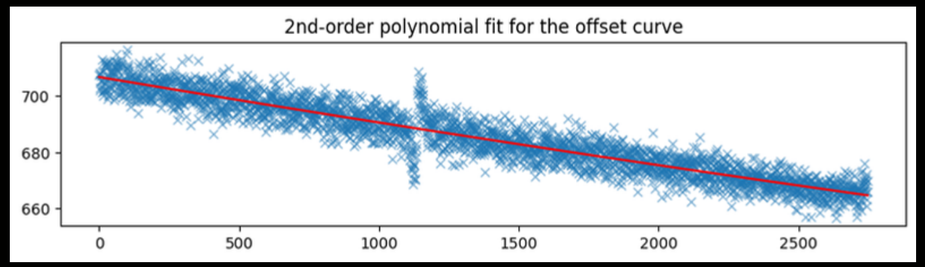


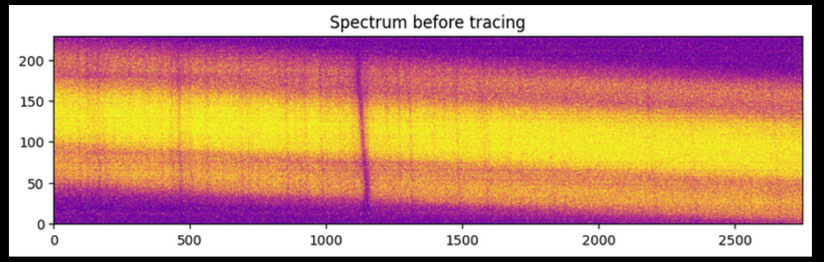
DATA & METHODS

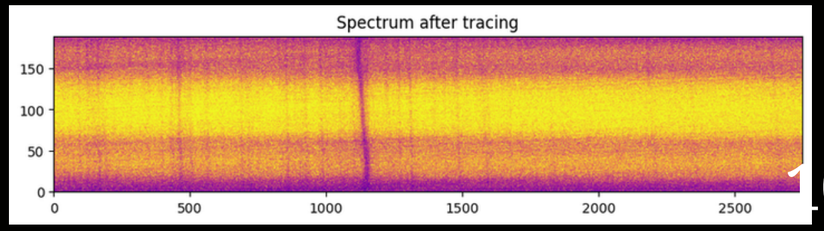
Data reduction



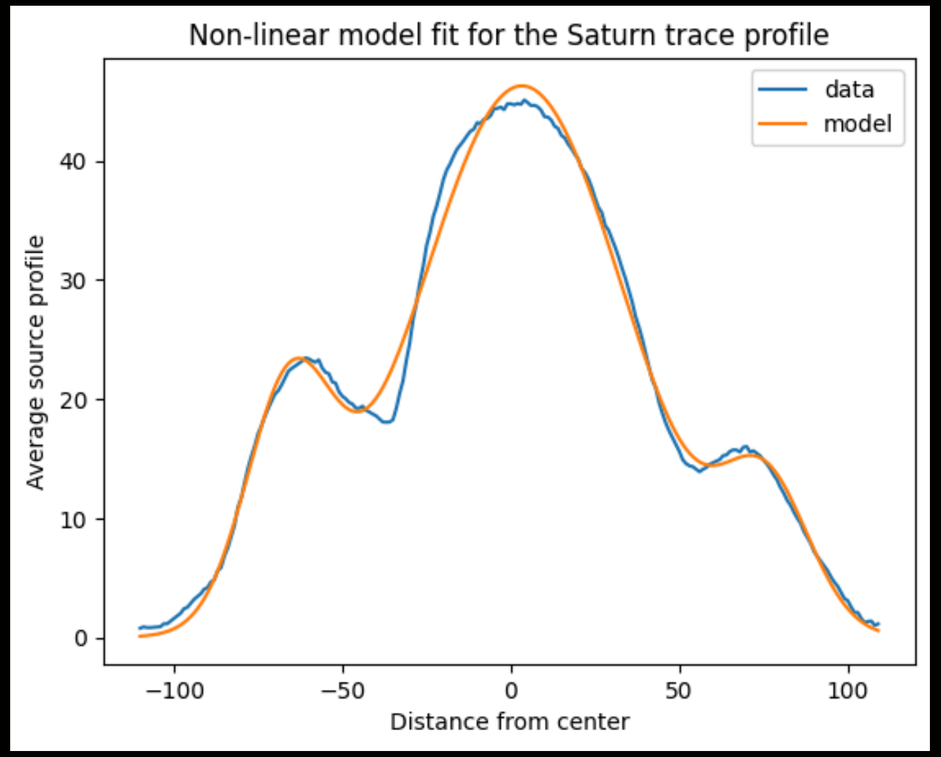
Tracing

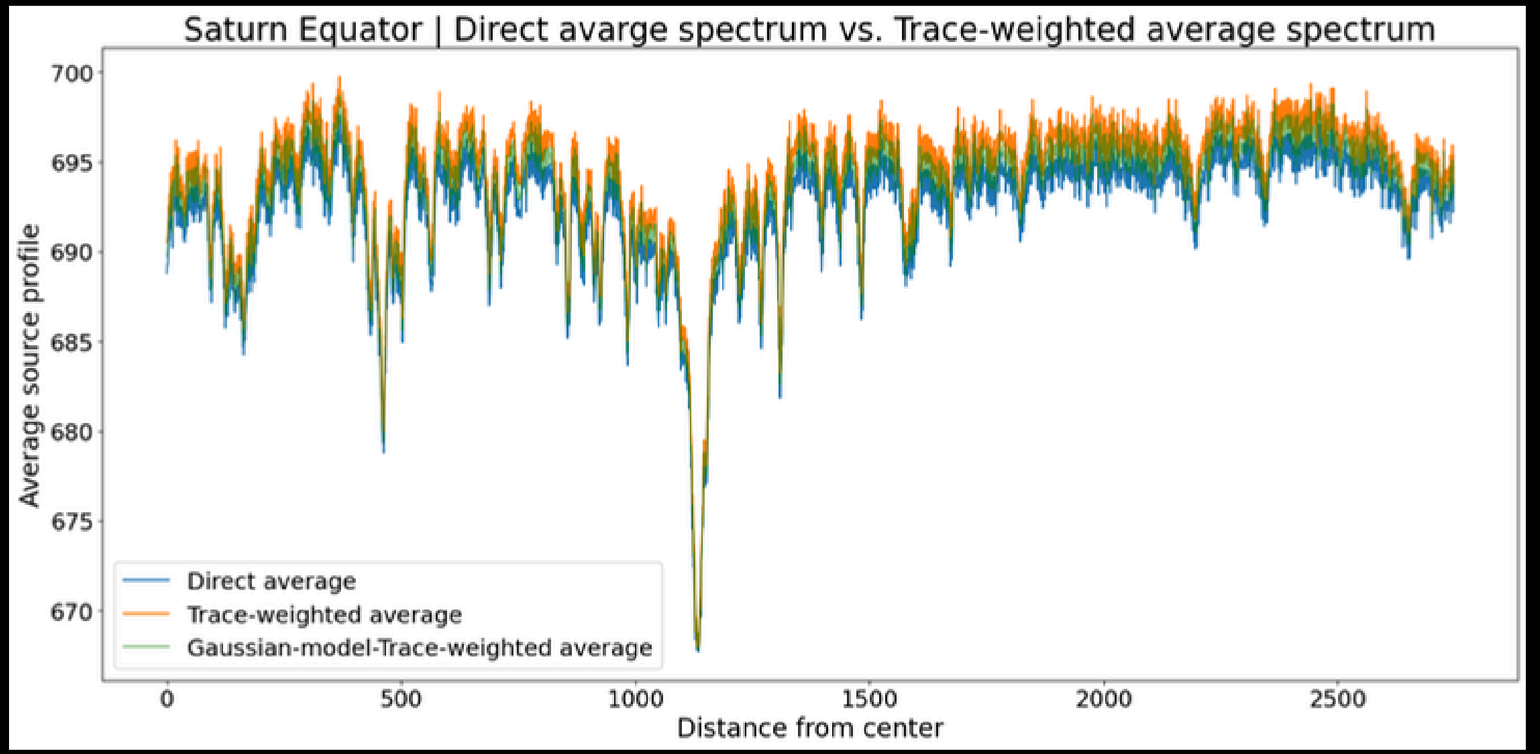






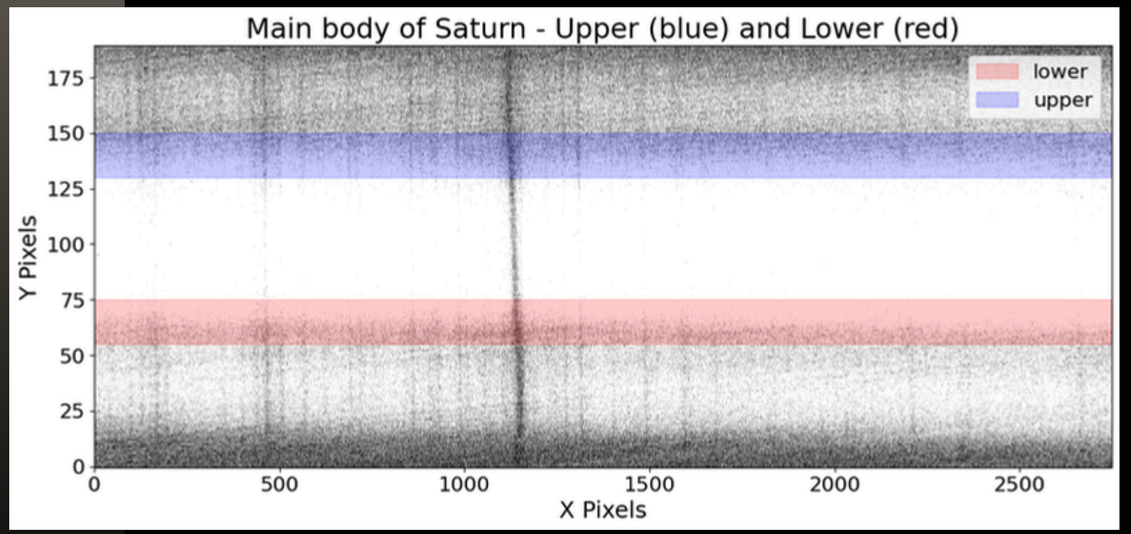
DATA & METHODS

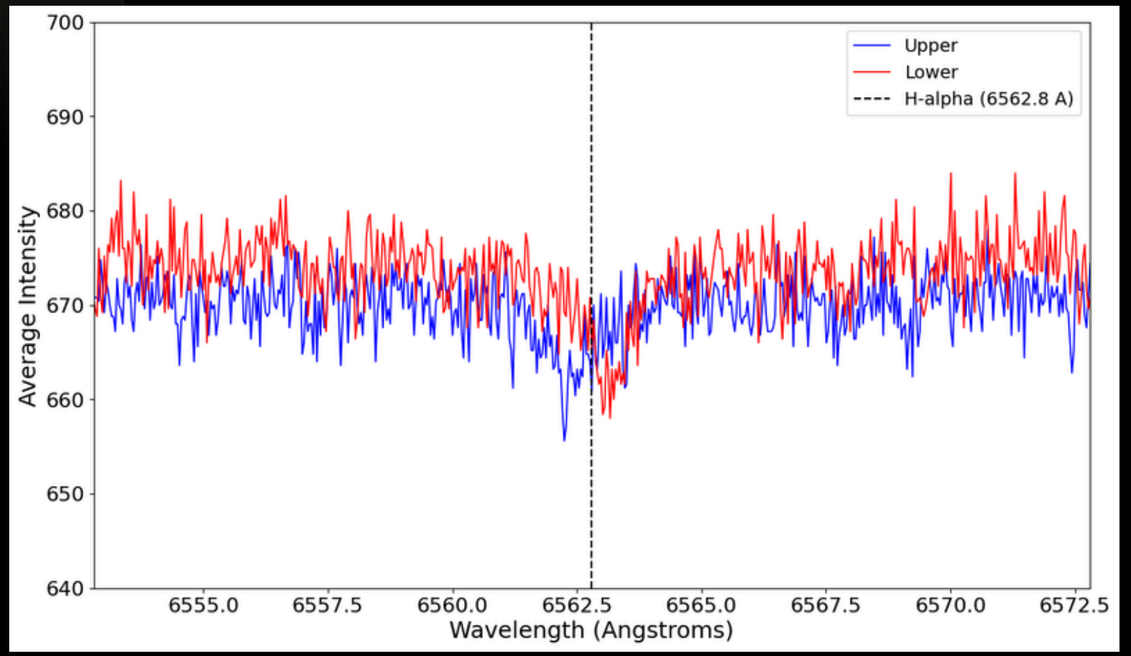




We computed a compound Gaussian fit for our trace profile.
Then, we compared the direct average spectrum and the trace-weighted average spectrum.

RESULT & DISCUSSION





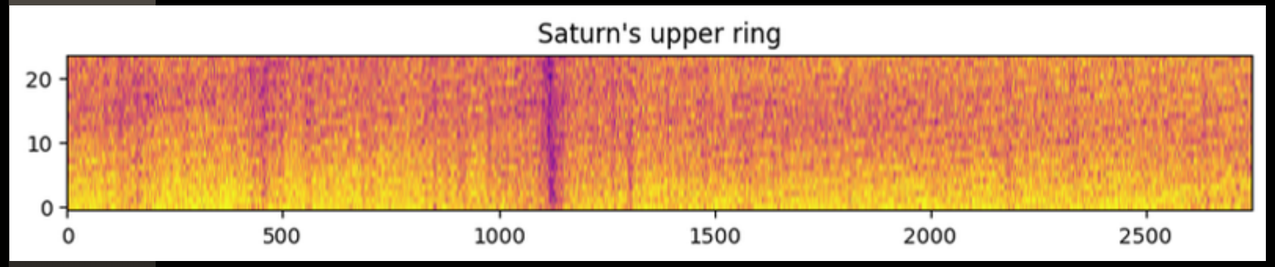
The rotational velocity is calculated by :

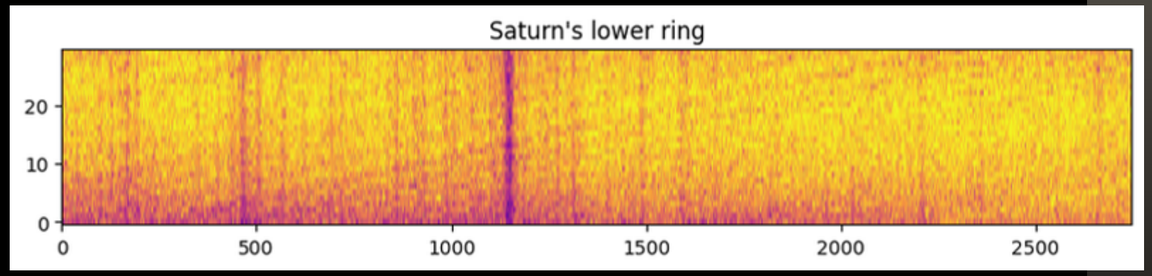
$$v_r = \frac{1}{4} \times c \times \frac{\lambda_{lower} - \lambda_{upper}}{6562.8}$$

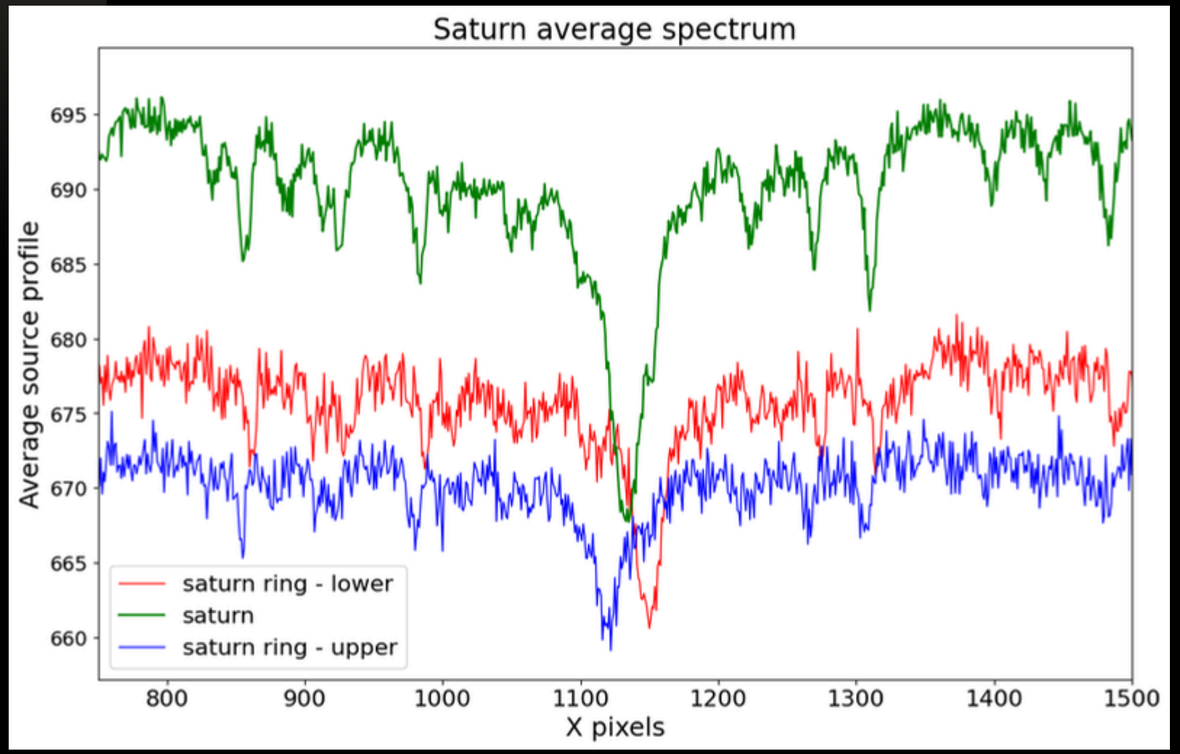
Region	Center wavelength (Angstrom)	FWHM (Angstrom)
Upper	6562.611	0.676
Lower	6563.347	0.873

We obtained the rotational velocity: v_{rot} = 9.36 km/s, which is 5.16% difference from the known result of 9.87 km/s

RESULT & DISCUSSION







The slant of H alpha is not visible in the ring region

However, their spectra indicate that the Saturn's Ring also rotates in the same direction as the main body.

CONCLUSION

Our project demonstrated that Doppler spectroscopy can be used to measure the rotational velocity of Saturn. Using observations from Quy Nhon Observatory, we processed the spectral data with Demetra and Python.

By analyzing the Doppler shift of the H-alpha absorption line, we estimated Saturn’s equatorial rotational velocity to be 9.36 km/s (5% agreement). We also found out that the Saturn ring also rotates with the planet.

For future work, we recommend improving slit alignment, analyzing additional spectral lines to increase precision, and studying Saturn’s rings with higher resolution to investigate their Keplerian rotation profile.

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[4]: Gurnett, D. A., et al. (2009). Discovery of a north-south asymmetry in Saturn’s radio rotation period. Geophysical Research Letters.

[5]: Desch, M. D., & Kaiser, M. L. (1981). Voyager measurement of the rotation period of Saturn’s magnetic field. Geophysical Research Letters.

THANK YOU SO MUCH

Q & A SESSION

